Section 8B Finding the Value of an Expression

When $x = 4$, the value of the expression $3x + 5$ is $3(4) + 5 = 12 + 5 = 17$.
When $n = 2$, the value of the expression $6n - 10$ is $6(2) - 10 = 12 - 10 = 2$

Example 1

If $x = 3$ and $y = 5$, find the value of each of the following expressions

(i) $4x + y$ (ii) $7x - 2y$ (iii) $6x + 2y - 7$

(i) $4x + y = 4(3) + 5 = 12 + 5 = 17$
(ii) $7x - 2y = 7(3) - 2(5) = 21 - 10 = 11$
(iii) $6x + 2y - 7 = 6(3) + 2(5) - 7 = 18 + 10 - 7 = 21$

Powers of *x*

As with numbers, $4^3 = 4 \times 4 \times 4$,
so also with variables, $x^3 = x \times x \times x$.
Similarly $a^2 = a \times a$ and $6a^3 = 6 \times a \times a \times a$

Example 2

Find the value of $3x^2 + 2x - 5$ when (i) $x = 2$ (ii) $x = 5$

(i) when $x = 2$; $3x^2 + 2x - 5 = 3(2)^2 + 2(2) - 5 = 12 + 4 - 5 = 11$
(ii) when $x = 5$; $3x^2 + 2x - 5 = 3(5)^2 + 2(5) - 5 = 75 + 10 - 5 = 80$

Exercise 8B

1. If $x = 2$ and $y = 3$, find the value of each of these:

(i) $4x$ (ii) $3y$ (iii) $2x + y$ (iv) $3x + 2y$
(v) xy (vi) $3xy$ (vii) $5x - y$ (viii) $6y - 4x$

2. If $a = 4$ and $b = 2$, find the value of:
(i) $a - b$ (ii) $2a + 3b$ (iii) $3a - 2b$ (iv) ab
(v) $3ab$ (vi) $2ab + 6b$ (vii) $2ab - 3a$ (viii) $2ab + 3a - b$

3. If $a = 1$, $b = 2$ and $c = 3$, find the value of:
(i) $2a + b$ (ii) $3ab - c$ (iii) $4abc + 3c$
(iv) $3bc - 4ab$ (v) $3abc - 2ac$ (vi) $5bc - 2ab$

4. If $x = 1\frac{1}{2}$ and $y = \frac{1}{2}$, find the value of:
(i) $4x + 2y$ (ii) $3x + y$ (iii) $2x - 4y$ (iv) $6x - 2y$

5. If $x = 4$, find the value of,
(i) $x^2 + 2x + 4$ (ii) $x^2 - 2x + 3$ (iii) $2x^2 + x - 6$

6. If $x = 1$ and $y = 2$. find the value of:
(i) x^2 (ii) $3x^2 + 2y$ (iii) $x^2 + 2y^2$ (iv) $6x^2 - 2y^2$

7. If $a = 3$ and $b = -2$, find the value of:
(i) b^2 (ii) $6b^2 - a^2$ (iii) $3a^2 + 2b^2$ (iv) $4a^2 - 3b^2$

8. Find the value of $3x^2 + 2x - 4$ when $x = 2$.

9. Find the value of $2x^2 - 3x + 6$ when $x = -2$.

10. If $A = 3.14\,r^2$, find the value of A when: (i) $r = 5$ (ii) $r = 10$

11. If $V = l^3$, find the value of V when $l = 4$.

12. Given that $s = 2t + 4t^2$, find the value of s when: (i) $t = 3$ (ii) $t = 5$ (iii) $t = 10$

13. If the speed of a car is given by the formula $v = 6 + 3a$, where v is the velocity and a is the acceleration, find the value of v when:
(i) $a = 4$ (ii) $a = 6$ (iii) $a = 8$.

14. If $h = 3n + 5$, find the value of h when: (i) $n = 4$ (ii) $n = 8$ (iii) $n = 12$.